

Internship Project Report: RetailPulse AI Sales Intelligence Platform

1. Executive Summary

RetailPulse is a comprehensive AI-driven retail analytics platform that combines sales intelligence, demand forecasting, customer segmentation, churn prediction, and inventory optimization. The platform uses cleaned transactional data from the Online Retail dataset to produce business-ready insights for revenue growth, customer retention, and stock planning.

2. Project Objectives

- Demand Forecasting: predict near-term sales using time-series modeling.
- Customer Intelligence: segment customers using RFM and K-Means clustering.
- Churn Prevention: identify inactive and high-risk customers with XGBoost classification.
- Inventory Optimization: rank SKUs with ABC analysis and calculate EOQ reorder quantities.
- Dashboard Delivery: provide a Streamlit interface and reproducible reporting pipeline.

3. Business Snapshot

Metric	Value
Total Revenue	\$8,911,407.90
Orders	18,532
Customers	4,338
Products	3,665
Average Order Value	\$480.87

4. Technical Methodology

Phase 1: Data Ingestion and ETL

The raw Excel data is loaded with Pandas, cancellations are removed, missing CustomerID values are filtered out, invalid quantities and prices are removed, and revenue is calculated at line-item level.

Phase 2: Feature Engineering

The pipeline creates invoice time features, RFM customer features, daily demand series, customer tenure, order cadence, and SKU-level demand measures.

Phase 3: Model Development and Validation

The project trains or computes dedicated models for forecasting, customer clustering, churn risk, ABC classification, and EOQ inventory recommendations.

Phase 4: Dashboard Integration and Deployment

The analytics layer is integrated into a Streamlit dashboard with custom styling, reproducible outputs, tests, and Docker metadata.

5. AI Techniques and Algorithms

5.1 Demand Forecasting

The forecasting module uses Prophet when available and otherwise falls back to a seasonal weekday baseline. The current run used Seasonal weekday baseline with a backtest MAPE of 18.22%.

5.2 Customer Segmentation: K-Means Clustering

RFM and behavioral features are scaled with StandardScaler and clustered into four business-friendly segments: Champions, Loyal, At Risk, and Hibernating.

Segment	Customers	Revenue Share
Loyal	3,228	67.6%
Champions	25	23.2%
Hibernating	1,083	6.5%
At Risk	2	2.8%

5.3 Churn Prediction: XGBoost Classifier

Churn is defined as inactivity longer than 90 days. The current model is XGBoost Classifier, with time-based holdout AUC-ROC of 0.72 and average precision of 0.61. It flags 612 customers as high risk. Recency and last-purchase fields are excluded from model inputs to avoid target leakage.

5.4 Inventory Optimization: ABC Analysis and EOQ

SKU revenue contribution is converted into A, B, and C priority classes. EOQ estimates optimal reorder quantity using annualized demand, ordering cost, and holding cost assumptions.

6. Technology Stack

- Programming Language: Python 3.12 deployment runtime
- Dashboard Framework: Streamlit with custom CSS
- Machine Learning: scikit-learn, XGBoost, optional Prophet-compatible forecasting
- Visualization: Plotly, Matplotlib, Seaborn
- Reporting: ReportLab PDF generation with pipeline artifacts
- Deployment: Dockerfile and requirements-based environment

6.1 System Architecture

The application follows a layered architecture: raw and processed transaction data feed a reusable Python analytics package; descriptive analytics, forecasting, clustering, churn scoring, and inventory optimization generate versionable tables; Streamlit provides the operational interface; ReportLab and Remotion produce reviewer-ready PDF and video deliverables; Docker and Render configuration provide a reproducible deployment path.

7. Model Outputs

Forecast Date	Predicted Revenue	Lower	Upper
2011-12-10	\$0	\$0	\$23,726
2011-12-11	\$22,299	\$0	\$46,025
2011-12-12	\$36,939	\$13,213	\$60,665
2011-12-13	\$48,320	\$24,594	\$72,046
2011-12-14	\$42,608	\$18,882	\$66,334
2011-12-15	\$55,003	\$31,277	\$78,729
2011-12-16	\$37,046	\$13,321	\$60,772

SKU	Description	ABC	EOQ	Reorder Point
23843	PAPER CRAFT , LITTLE BIRDIE	A	3,899	3,032
22423	REGENCY CAKESTAND 3 TIER	A	623	464
85123A	WHITE HANGING HEART T-LIGHT HOLDER	A	2,226	1,375
85099B	JUMBO BAG RED RETROSPOT	A	2,990	1,729
23166	MEDIUM CERAMIC TOP STORAGE JAR	A	4,993	2,917
POST	POSTAGE	A	196	117
47566	PARTY BUNTING	A	1,106	572
84879	ASSORTED COLOUR BIRD ORNAMENT	A	2,866	1,324

8. Output Visuals

Fig 1: Dashboard-Enterprise-Overview

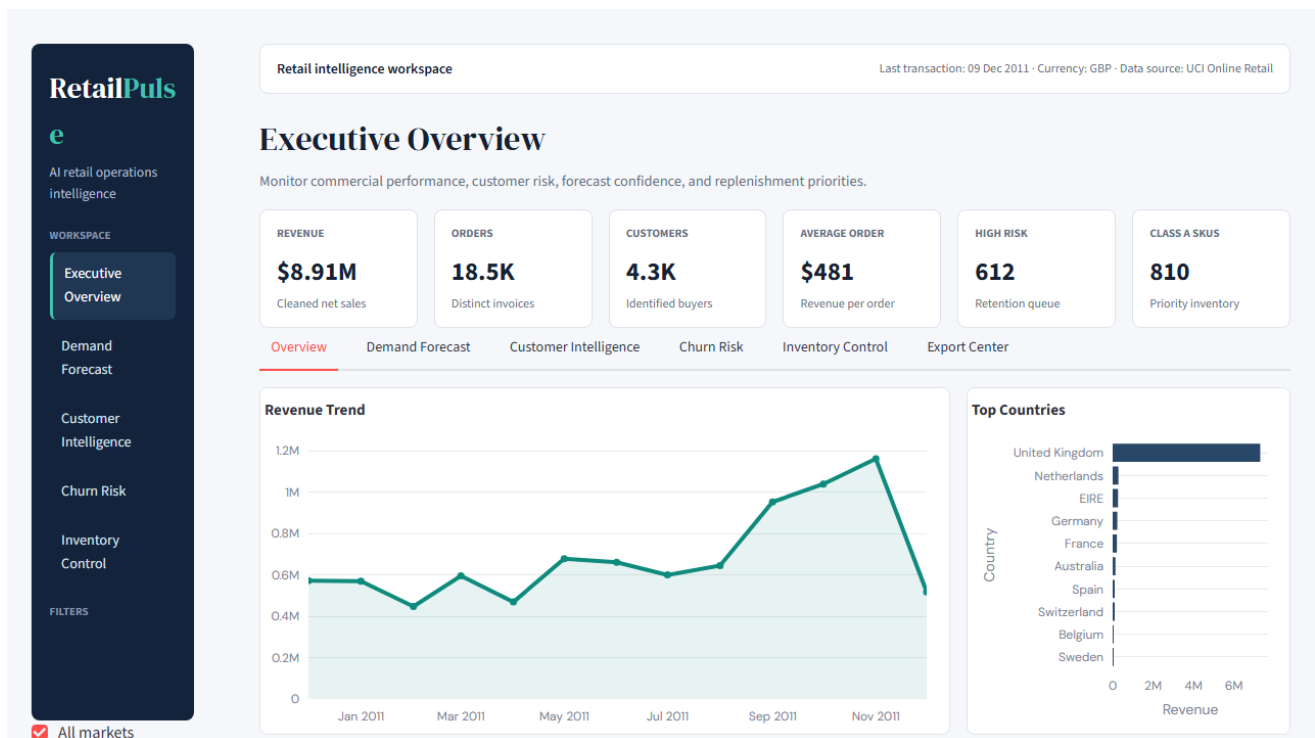


Fig 2: Monthly Revenue

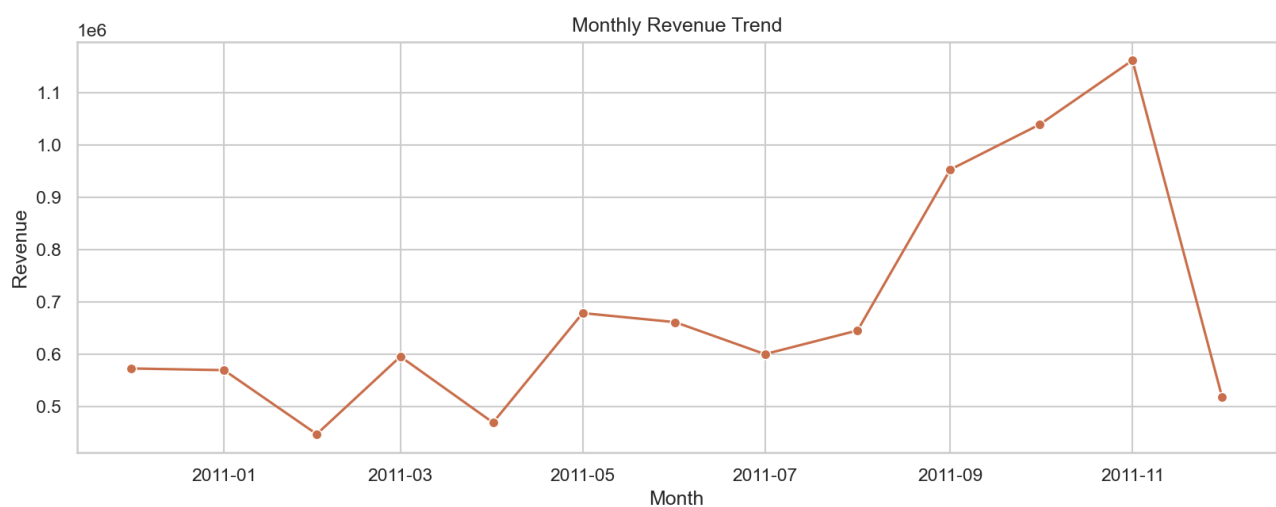


Fig 3: Top Countries

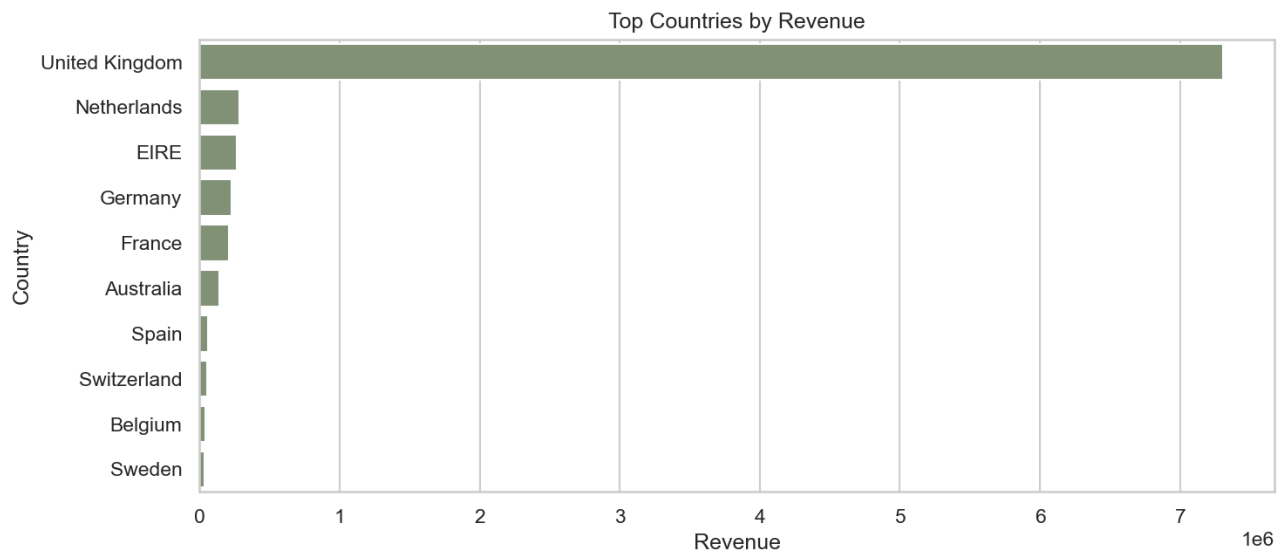


Fig 4: Customer Segments

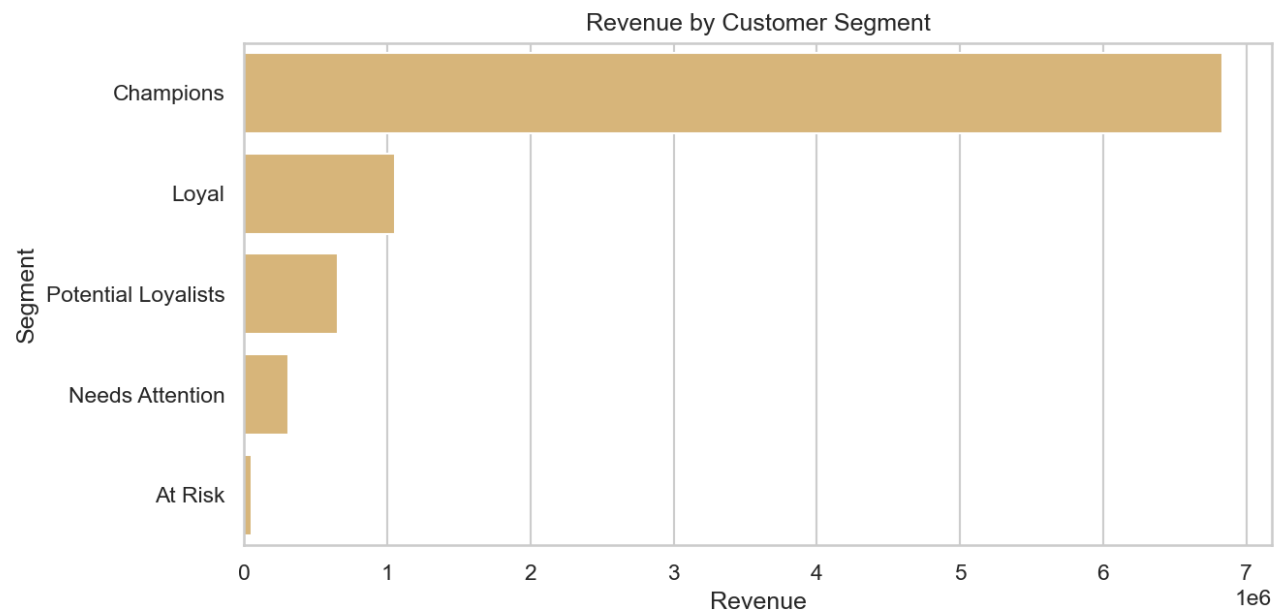


Fig 5: Demand Forecast

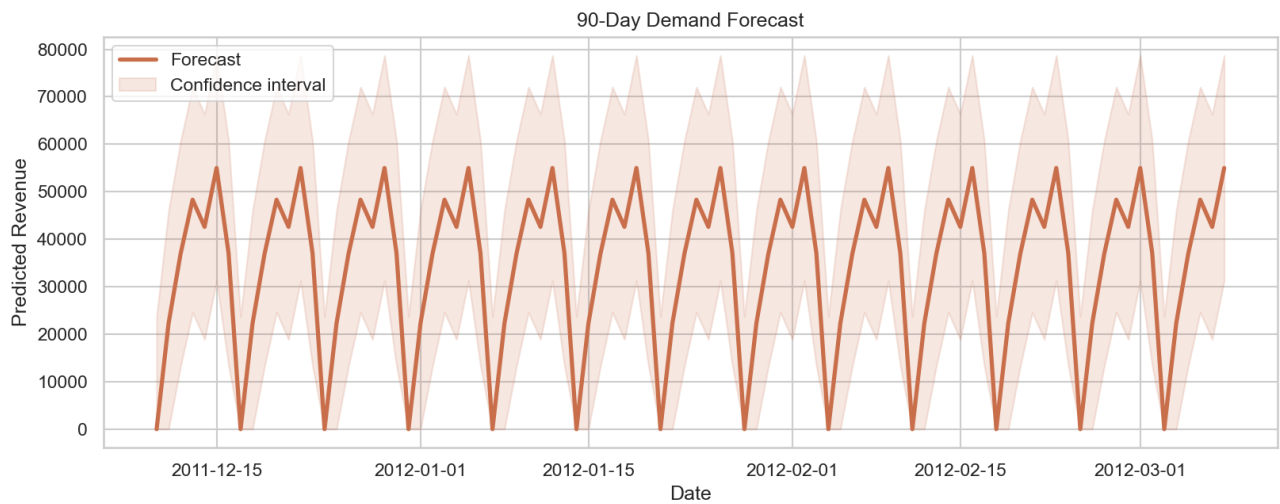


Fig 6: Churn Risk

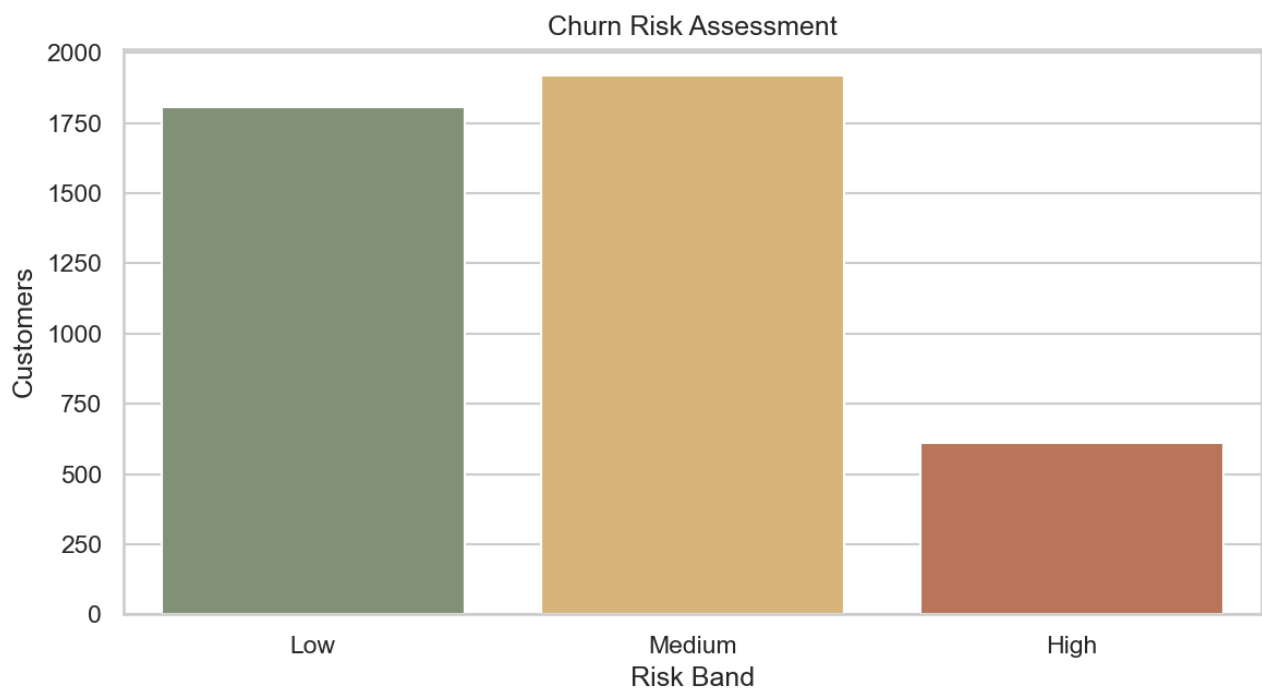
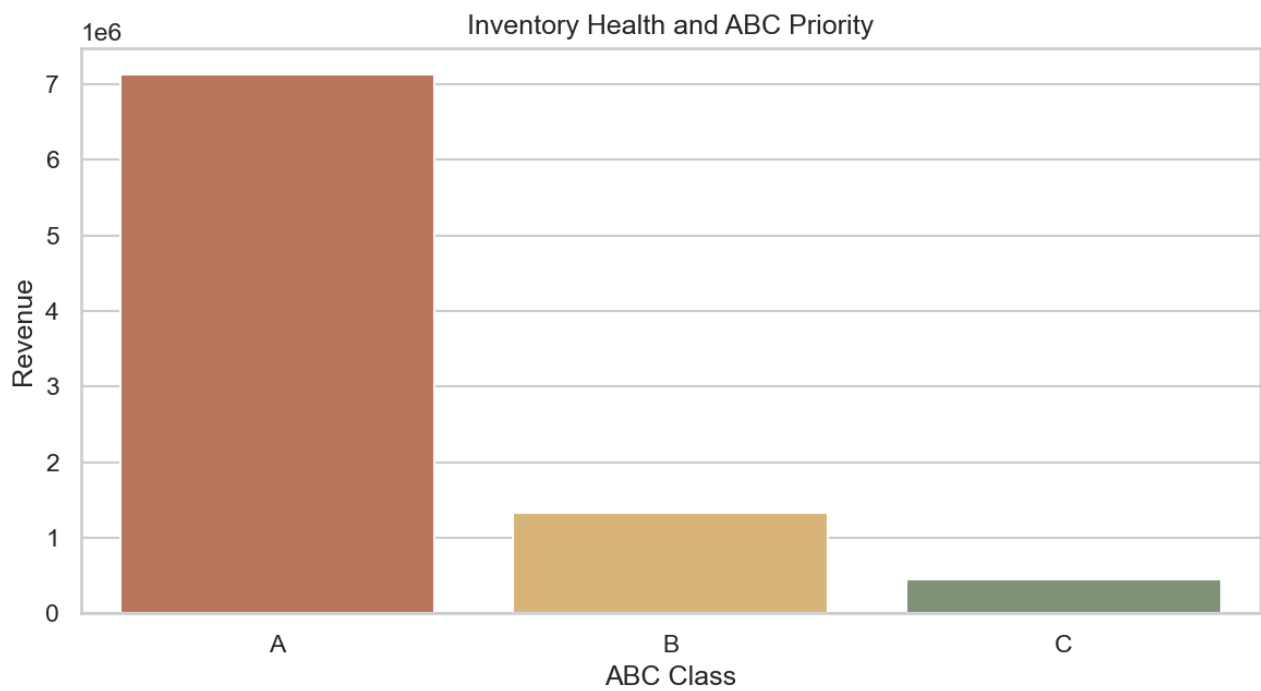


Fig 7: Inventory Abc



9. Conclusion

RetailPulse now demonstrates a complete applied retail AI workflow: cleaned transactional data, advanced customer and inventory modeling, predictive forecasting, operational recommendations, a polished dashboard, and a comprehensive PDF report suitable for internship or portfolio submission.

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